

HP ProLiant AMD-based 100-series G7 servers

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Abstract

This technology brief describes the key technologies implemented in HP ProLiant 100-series G7 servers based on AMD™ processors. As of this writing, the AMD-based 100-series G7 server platform is limited to the ProLiant DL165. A link to the QuickSpecs for this server is listed in the “For more information” section at the end of this technology brief.

Introduction

The HP ProLiant DL165 G7 server platform is a high performance, low cost, ultra-dense rack server node for compute and I/O intensive environments. Several key technologies are included:

- AMD Opteron™ eight- and twelve-core 6100 Series processors, including high efficiency (HE) processor options
- Double Data Rate-3 (DDR3) memory DIMMs
- Lights-Out 100i (LO 100i) and management and deployment tools including SNMP management agents and the Easy Setup CD
- Input/output (I/O) technologies such as PCIe generation 2 (PCIe 2.0) and faster Smart Array controllers that incorporate common form factor components
- B110i software RAID with optional Serial Advanced Technology Attachment (SATA) Hot Plug technology
- Common Slot power supplies to provide the required amount of power and improve power efficiency

HP servers and balanced architecture

HP designs cost-competitive, power-efficient servers that use a balanced architecture to address performance requirements and provide value.

HP servers achieve a balanced architecture through superior engineering of fundamental elements such as mechanical infrastructure, power, cooling, processor, memory, IO devices, storage, boot, networking, and interconnecting components. A balanced architecture includes the following:

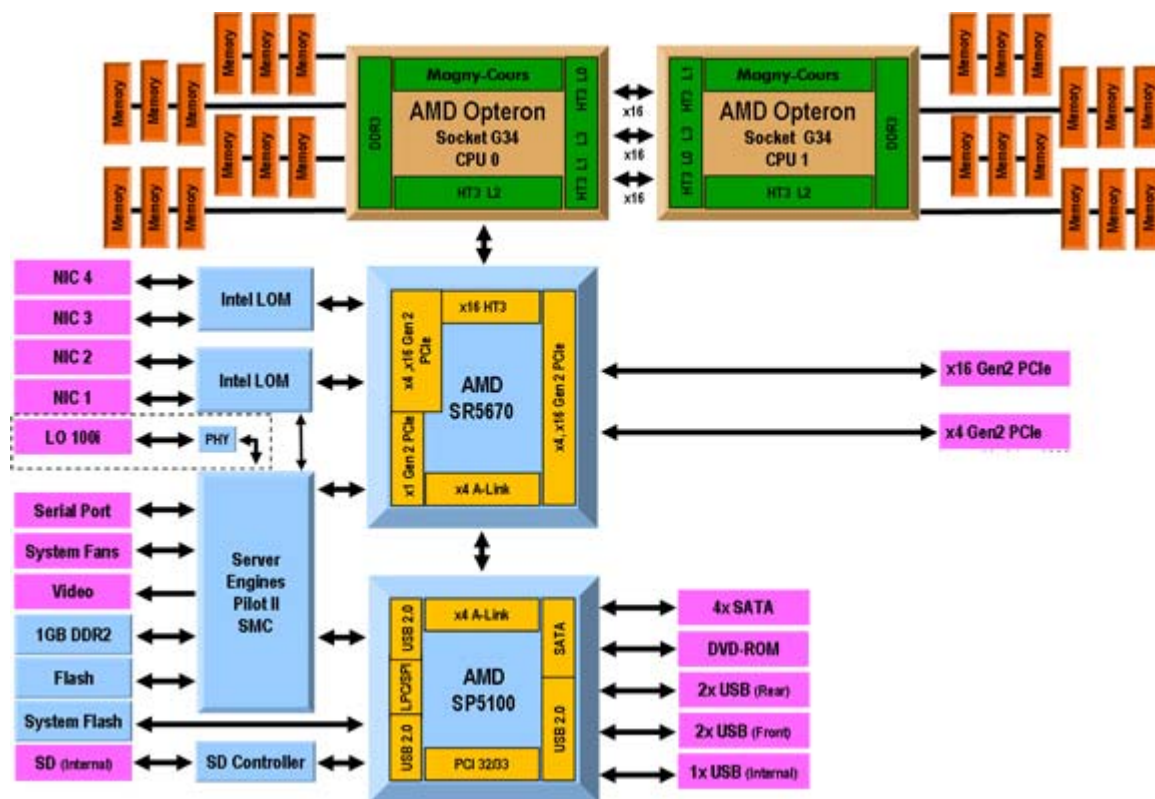
- Compute capability, processor core count, cache size per processor, and processor socket count
- Low-latency processor-to-memory bandwidth commensurate with core count
- Memory footprint and capacity that maximizes bandwidth and capacity with power efficiency and performance without compromising quality or reliability
- Application-appropriate I/O devices
- Closely-coupled and balanced processor-to-memory and processor-to-I/O ratios
- Mechanical design that ensures optimum levels of cooling, stability, and serviceability through space-efficient, modular partitioning across the server

By designing a balanced architecture, HP ensures that all subsystems can be used effectively under a broad range of applications and workloads. For example, increasing memory capacity asymmetrically will not increase performance as effectively as distributing the same amount of memory across processors and IO devices. Inefficient memory distribution yields diminishing returns on power consumption and cost. A virtual machine (VM), for example, benefits from memory closely coupled to the processor responsible for that VM. Furthermore, a server needs to have appropriate levels of I/O bandwidth and CPU capabilities to ensure that memory can be used effectively by every VM.

Processor technology

The HP G7 AMD-based 100-series servers use the AMD Opteron 6100 series eight- and twelve-core processors, allowing up to 24 cores in a 2-processor, 1 U platform. These processors are based on AMD's 45 nanometer (nm) process and have a core standard wattage of 75W Average CPU Power (ACP). In addition, these processors use Direct Connect™ architecture 2.0, HyperTransport™ 3.0, HT Assist™, and 8x ECC error correction. The 6100 series processors fit into the G34 socket architecture. They feature DDR3 memory, four 64 bit memory channels per processor, the AMD SR5670 chipset, and quad HyperTransport (HT) links as shown in Figure 1. The processors operate at speeds of up to 2.4 GHz with eight cores and 2.3 GHz with twelve cores, and share a 12 MB L3 cache. The AMD Opteron 6100 series processor includes performance-optimized power and thermal controls. See the [Power and thermal technologies](#) section for more information.

Figure 1. The DL165 G7 server architecture using the AMD 6100 series processor and chipset

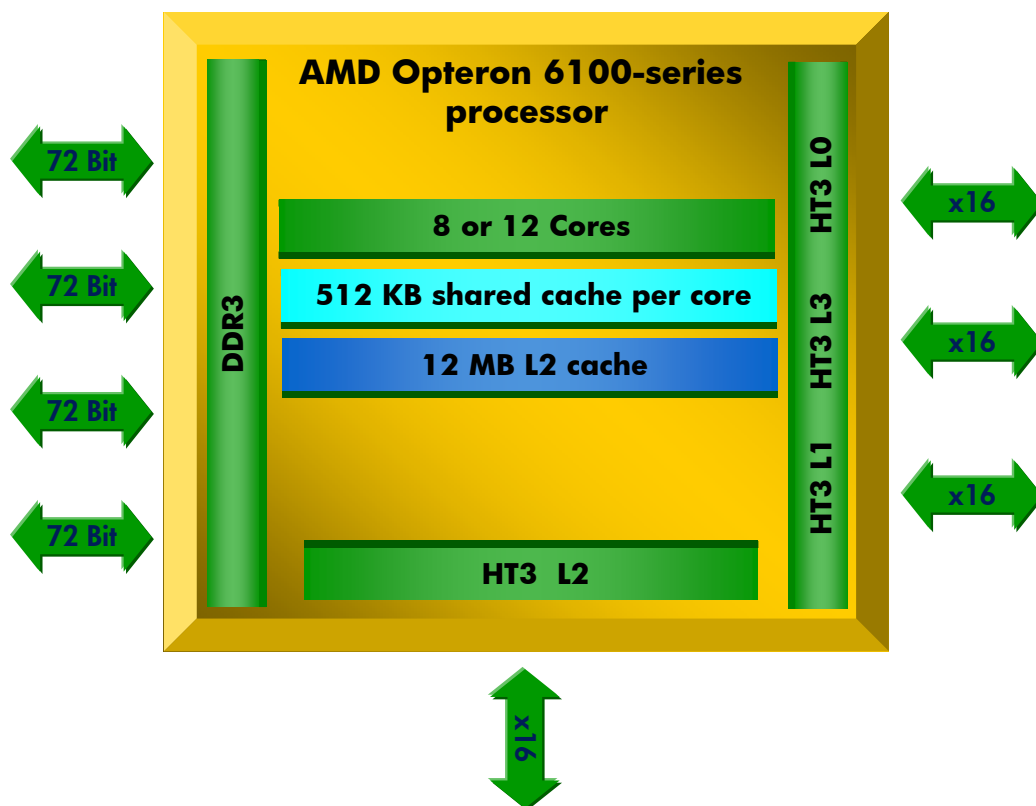


Direct Connect architecture 2.0

Unlike front-side bus architecture, Direct Connect integrates the memory controller into the processor and directly connects CPUs to the I/O subsystem and other processors (Figure 2). Direct Connect architecture uses direct communication HyperTransport links between CPUs, between CPU and I/O, and between CPU and memory. Direct Connect architecture currently scales up to 12 cores to provide

superior memory and I/O capability, near native virtualization performance, and a range of power bands¹ that place a priority on low power consumption.

Figure 2. Block diagram of Direct Connect 2.0 architecture in the AMD 6100 series processors



HyperTransport technology

HyperTransport is a point-to-point interconnect that is designed to connect the processors directly and to connect each processor to its dedicated memory banks, as well as to other I/O chipsets.² Compared to a shared, parallel front-side bus, the advantages of HyperTransport include no overhead for bus arbitration and easier signal integrity maintenance, which results in a scalable, high-bandwidth architecture.

Each 16-bit (2-byte) HyperTransport link is double-pumped to perform two data transfers per clock cycle. From HyperTransport 1.0 (HT1) in 2001 to HyperTransport 3.0 in 2008, the maximum clock speed and transfer rate increased from 800 MHz (1.6 MT/s³) to a maximum of 2.6 GHz (5.2 GT/s) in each direction.

¹ Power bands refer to a new metric developed by AMD to reflect power consumed by the processor and its integrated memory controller during peak workloads. This metric is based on AMD's measurement of Average CPU Power (ACP). For more information on ACP, see the whitepaper at www.amd.com/us-en/assets/content_type/white_papers_and_tech_docs/43761C_ACP_WP.pdf

² HyperTransport Technology was invented at AMD with contributions from industry partners and is managed and licensed by the HyperTransport Technology Consortium, a Texas non-profit corporation.

³ MT/s, or megatransfers per second, equals the speed of the link in millions of cycles per second times the number of transfers per cycle.

The G34 socket infrastructure features four, 16-bit HT3 links operating at up to 6.4 GT/s per link. Of the four HT3 links, three are shared processor to processor links and one link on each processor is available for I/O communication.

HT Assist

HT Assist reduces cache coherence traffic on the HT links. By tracking where data is stored in cache and guiding the processor directly to the caches of other processors, HT Assist reduces cache probe traffic between processors, especially in 4-socket servers. Therefore, HT Assist results in faster queries that can increase performance for cache-sensitive applications such as database, virtualization, and compute-intensive applications.

Memory technologies

In the AMD Opteron 6100-series architecture, the memory controller is integrated into the processor chip to optimize memory performance and bandwidth per CPU. The memory controller reduces latency inherent in front side bus architectures by eliminating the bus contention between memory and I/O cycles. AMD's memory bandwidth increases as processors are added to a configuration, unlike legacy designs that scale poorly because access to main memory is limited by external Northbridge chips. HP designed specific BIOS and BIOS Setup Utility (BSU) functions to manage memory configurations, letting the customer optimize configurations for maximum performance while reducing power consumption and cooling requirements. These management options also include memory protection and latency reduction. All of these options are examined later in this section.

DDR3 DIMM choices

HP ProLiant G7 100-series servers with Opteron 6100 series processors support double data rate (DDR3) dual inline memory modules (DIMMs). DDR3 has several key enhancements including an 8-bit prefetch buffer for storing data before it is requested. By comparison, DDR-2 has a 4-bit buffer. For DDR3, the data signal rate can increase to 1333 Megatransfers per second (MT/s). While this is commonly referred to as having a speed of 1333 MHz, the maximum clock speed for the DIMMs is actually 667 MHz and the signal is double-pumped to achieve the 1333 MT/s data rate. DDR3-1333 DIMMs can operate at clock speeds of 667 MHz, 533 MHz, and 400 MHz with corresponding data rates of 1333, 1066, and 800 MT/s.

HP DDR3 DIMM modules incorporate an integrated thermal sensor that signals the chipset to throttle memory traffic to the DIMM if its temperature exceeds a programmable critical trip point. Using the data from these thermal sensors, ProLiant AMD 100-series G7 servers can reduce fan speed when memory is idle, effectively reducing power consumption. The BIOS in ProLiant G7 servers verifies the presence of the thermal DIMM sensor during POST. Some third-party DIMMs may not include this thermal sensor. If it is absent, a POST message will warn that the DIMM does not have a thermal sensor, and the fans will be forced to run at higher speeds (requiring more power).

DDR3 is available as both Unbuffered Dual In-line Memory Modules (UDIMMs) and Registered (buffered) Dual In-line Memory Modules (RDIMMs). Both RDIMMs and UDIMMs support error correcting code (ECC).

There are three types of standard voltage DDR3 available for ProLiant 100 series G7 servers:

- PC3-8500R- DDR3 (RDIMM, ECC compliant)—1333 or 1066 MT/s data rate depending on memory configuration.
- PC3-10600R DDR3 (RDIMM, ECC compliant)—1333 or 1066 MT/s data rate depending on memory configuration.
- PC3-10600E DDR3 (UDIMM, ECC compliant)—1333 MT/s data rate

Low-voltage DDR3

Customers can take advantage of the HP low voltage (LV) DDR3 memory option. LV memory can operate at 1.35 V, reducing power and cooling requirements. Some LV DIMM configurations can affect performance. The difference occurs in 2 DIMM per channel (DPC) configurations with dual-rank DIMMs. In these configurations, the memory bus runs at 1.5 V with 2 DPC at 1333 MT/s, or 1.35 V with 2 DPC at 1066 MT/s. This results in a 20% reduction in bandwidth for the lower voltage. In all other HP LV configurations, the LV data rate is the standard 1.5V data rate. The BIOS determines the DIMM population from the system and the operating voltage and data rate capability from the DIMMs. The BIOS then sets the data rate based on that information.

NOTE:

Although the bandwidth reduction from DDR3-1333 to DDR3-1066 is 20%, the measured reduction in throughput is 10%

Single rank DIMMs configured at 1, 2, and 3 DPC; and dual rank DIMMs configured at 1 and 3 DPC run at the 1333 MT/s data rate at both voltages.

ProLiant G6 server models with the AMD 6100-Series processor support LV memory:

- PC3L-10600R—1333 1066 or 800 MT/s data rate, depending on memory configuration.

DIMM configuration guidelines

ProLiant AMD-based 100-series G7 servers support 3 DIMMs per channel for up to 24 DDR3 DIMMs. DDR3 DIMM speeds will vary depending on number of DIMMs per channel. Consult the server QuickSpecs to determine DIMM speeds for given configurations. Administrators can configure Opteron-based ProLiant 100-series G7 servers using either RDIMMs or UDIMMs but RDIMM and UDIMM memory cannot be mixed within a single server. Low voltage DIMMs can provide up to 10% DIMM power savings. Low and standard voltage DIMMs are compatible; systems automatically adjust voltage based on DIMMs installed.

For detailed memory configuration rules and guidelines, please use the Online DDR3 Memory Configuration Tool at www.hp.com/go/ddr3memory-configurator

Memory management technologies

Memory interleaving on the AMD Opteron 6100-series processors can occur between the processor memory banks, memory channels, and between processor nodes in a multiprocessor system. All three memory interleaving technologies are supported on the ProLiant AMD 100-series G7 servers. These technologies are independent of each other and can operate simultaneously.

Memory bank interleaving

When memory bank interleaving is engaged, data is routed alternately to memory banks through the common memory channel connecting the DIMM banks and the integrated memory controller. However, memory bank interleaving does increase the probability that more DIMMs need to be kept in an active state (requiring more power) since the memory controller alternates between memory banks and therefore between DIMMs.

Memory bank interleaving is automatically enabled on a processor node under the following conditions:

- Two single-rank DIMMs per channel result in two way bank interleaving
- Two dual-rank DIMMs per channel result in four way bank interleaving

- Two quad-rank DIMMs per channel results in eight way bank interleaving
- Two dual-rank DIMMs and one quad-rank DIMM results in eight way bank interleaving (in servers using three DIMMs per channel)

Memory channel interleaving

With memory channel interleaving, data is transferred by means of alternate routing through the two available memory channels. The result is that when the memory controller needs to access a block of logically contiguous memory, the requests are distributed more evenly across the two channels rather than potentially stacking up in the request queue of a single channel. This alternate routing decreases memory access latency and increases performance. As with memory bank interleaving, memory channel interleaving increases the probability that more DIMMs need to be kept in an active state.

Memory channel interleaving is always active on the AMD 6100-series processor.

Memory node interleaving

With node interleaving, memory can be interleaved across any subset of nodes in the multiprocessor system. Node interleaving breaks memory into 4 KB addressable entities. Addressing starts with address 0 on node 0 and assigns sequential addresses through address 4095 to node 0, addresses 4096 through 8191 to node 1, addresses 8192 through 12287 to node 2, and addresses 12888 through 16383 to node 3. Address 16384 is assigned to node 0, and the process continues until all memory has been assigned in this fashion. An application that uses a common allocation thread will benefit from node interleaving.

Memory node interleaving is disabled by default. Administrators can activate node interleaving using the BSU. Node interleaving can only be configured if the memory footprint for both processors is identical.

8X error correction

In AMD Opteron 6100-series processors, the memory controller supports error correction circuitry (ECC) for both x4 and x8 DIMMs.

I/O technologies

ProLiant 100-series G7 servers incorporate PCI Express, Serial-Attached SCSI (SAS), and SATA I/O technologies. This server architecture lets administrators add PCI Express-compliant expansion cards to the system. SAS is a serial communication protocol for direct-attached storage devices such as SAS and SATA hard drives.

PCI Express technology

All ProLiant G7 servers support the PCIe 2.0 specification. PCIe 2.0 has a per-lane signaling rate of 5 Gb/s—double the per-lane signaling rate of PCIe 1.0. PCIe 2.0 is completely backward compatible with PCIe 1.0. A PCIe 2.0 device can be used in a PCIe 1.0 slot and a PCIe 1.0 device can be used in a PCIe 2.0 slot. Table 1 shows the level of interoperability between PCIe cards and PCIe slots.

Table 1. PCIe device interoperability

PCIe device type	x4 Connector x4 Link	x8 Connector x4 Link	x8 Connector x8 Link	x16 Connector x8 Link	x16 Connector x16 Link
x4 card	x4 operation	x4 operation	x4 operation	x4 operation	x4 operation
x8 card	Not allowed	x4 operation	x8 operation	x8 operation	x8 operation
x16 card	Not allowed	Not allowed	Not allowed	x8 operation	x16 operation

HP Smart Array and SAS/SATA technology

The newest serial PCIe 2.0-capable Smart Array controllers use SAS technology, a point-to-point architecture in which each device connects directly to a SAS port rather than sharing a common bus as parallel SCSI devices do. Point-to-point links increase data throughput and improve the ability to locate and fix disk failures. More importantly, SAS architecture solves the parallel SCSI problems of clock skew and signal degradation at higher signaling rates.⁴

The latest Smart Array controllers are compatible with SATA technology and include the following features to enhance performance and maintain data availability and reliability:

- **SAS and SATA compatibility** — SAS-2 compliance lets administrators deploy and manage both SAS arrays and SATA arrays. Smart Array configuration utilities help administrators configure arrays correctly so that data remains available and reliable.
- **SAS wide port operations** — Wide ports contain four single lane (1x) SAS connectors and the cabling bundles all four lanes together. SAS wide ports enhance performance by balancing SAS traffic across the links. In addition, wide ports provide redundancy by tolerating up to three physical link failures while maintaining the ability to communicate with the disk drives. The tolerance for link failures is possible because wide port connections are established from Phy⁵ to Phy, and multiple, simultaneous connections to different destinations are possible. The most common use of these wide links is to a JBOD or to an internal server expander connecting to large numbers of drives. No special configuration is required for this functionality.
- **SAS expanders** — Low-cost, high-speed switches called expanders can combine multiple single links to create wide ports and increase available bandwidth. SAS expander devices also offer higher system performance by expanding the number of hard drives that can be attached to an HP Smart Array controller. SAS expanders are an aggregation point for large numbers of drives or servers providing a common connection. By cascading expanders, administrators can chain multiple storage boxes together. For more information on the HP SAS Expander Card, go to <http://h18004.www1.hp.com/products/servers/proliantstorage/arraycontrollers/sas-expander/index.html>.

SAS-2

SAS-2 and PCIe 2.0 are among the technologies responsible for a significant increase in performance over past generations of Smart Array controllers. The second-generation SAS (SAS-2) link speed⁶ of 6.0 Gb/s is double the SAS-1 transfer rate. Operation at SAS-2 link speeds requires SAS-2 compliant hard drives. SAS-2 eliminates the distinction between fanout and edge expanders by replacing them with self-configuring expanders. SAS-2 enables zoning for enhanced resource deployment, flexibility, security, and data traffic management. SAS-2 is also backward compatible with SAS-1.

Beginning with HP product releases in the first quarter of 2009, Smart Array controllers are SAS-2 capable. In fully supported controllers, 6-Gb/s SAS technology allows Smart Array controllers to deliver peak data bandwidth up to 600 MB/s per physical link in each direction. SAS devices are capable of sending and receiving data simultaneously across each physical link (full duplex mode). When running full duplex, 6-Gb/s SAS technology can deliver peak data bandwidth up to 1200 MB/s.

⁴ For more information about SAS technology, refer to the HP technology brief titled "Serial Attached SCSI storage technology" available at <http://h20000.www2.hp.com/bc/docs/support/SupportManual/c01613420/c01613420.pdf>.

⁵ The mechanism that contains a transceiver which electrically interfaces to a physical link. Phy is a common abbreviation for the physical layer of the OSI model.

⁶ The Serial Attached SCSI - 2 (SAS-2 or SAS 2.0) is a draft standard, and is the product of the Technical Committee 10t Organization. SAS 2.0 is second generation of SAS and is based upon SAS - 1.1. The SAS-2 specification is available from the 10t website, <http://www.t10.org>.

The SAS-2 specification is compatible with both Serial SCSI and Serial ATA protocols for communicating commands to SAS and SATA devices. SAS-2 compliant controllers are fully compatible with 1.5-Gb/s and 3.0-Gb/s SATA technology.

For the most up-to-date listing of HP Smart Array controllers that support the SAS-2 specification, see the Smart Array controller matrix: <http://www.hp.com/products/smartarray>

HP Smart Array controllers based on PCIe 2.0

The Smart Array PCIe 2.0-based controllers are modular solutions with a common form factor, hardware, and firmware. Any of the ProLiant 300-series G6 and G7 servers can use PCIe 2.0-based controllers. With the exception of the DL320 G6 and ML330 G6 servers, the ProLiant 300-series G6 and G7 servers incorporate embedded HP Smart Array P410i/Zero Memory Controllers (RAID 0/1/1+0). As a standard entry level RAID, HP designed the Smart Array 410i with a unique Zero Memory RAID (ZMR) capability. Administrators can choose the cache size and can choose to include the battery backed write cache (BBWC) or the flash-backed write-cache (FBWC). These options allow users to upgrade from ZMR to 512 MB BBWC or 1GB FBWC.

Battery backed write cache

The BBWC system continues to be an option for capacity expansion (adding one or more physical disks to an existing array). The Smart Array controller recalculates parity and balances the data across all the disks. During the expansion, the BBWC preserves data and logical structures on the array. The HP 650 mAh P-Series battery extends battery life up to 48 hours before recharging becomes necessary.

Flash-backed write cache

HP introduced the flash-backed write-cache (FBWC) system in the fourth quarter of 2009. The FBWC uses NAND⁷ flash devices to retain cache data and super-capacitors (Super-caps) instead of batteries to provide power during a power loss. The FBWC offers significant advantages over the HP Battery-backed write-cache (BBWC) system. Since the FBWC writes the contents of memory to flash devices, there is no longer a 48-hour battery life limitation and the data will be posted to the disk drive on the next power up.

The FBWC DDR2 mini-DIMM cache module is specifically designed for the present generation of PCIe2.0, SAS-based Smart Array controllers based on the PMC PM8011 SAS SRC 8x6G RAID on a chip (RoC). The primary FBWC components consist of the cache module, Super-caps with integrated charger, and RoC located on the system board.

At the time of this writing, the FBWC cache is supported on the Smart Array P410, P410i, P411, P212, P812, and P712m.

For more information on the flash-backed write cache, see the “HP Smart Array Controller technology brief” at

<http://h20000.www2.hp.com/bc/docs/support/SupportManual/c00687518/c00687518.pdf>

Zero Memory RAID

Using Zero Memory RAID (ZMR), administrators can create a RAID 0-1 configuration without additional memory. Smart Array P410, P411, and P212 controllers include ZMR. The P212 controller does not include ZMR on the external connector. ZMR supports up to eight drives in Zero Memory Mode, or seven drives and one tape drive. ZMR mode does not support Modular Smart Array (MSA) products. ZMR does not include caching; however, all systems can be upgraded to a BBWC or FBWC memory module that can significantly increase performance.

⁷ Non-volatile semiconductor memory that can be electronically erased and reprogrammed. No power is needed to maintain data stored in the chip

NOTE:

Smart Array Advanced Pack is not available on Zero Memory configurations.

Software RAID

HP has developed a software RAID solution based on the Smart Array firmware. The Smart Array B110i SATA Software RAID supports the Array Configuration Utility (ACU), ACU-CLI (command line interface), Simple Network Management Protocol (SNMP) agents, and Web-Based Enterprise Management (WBEM) providers.

The B110i features an OS-specific driver from HP that uses the embedded ICH10R controller. It can use RAID 0, 1, and 1+0 and supports a maximum of two logical drives. The B110i supports up to four 1.5Gb or 3Gb SATA drives. Because it is based on the Smart Array firmware, you can migrate drives to a hardware-based Smart Array controller in a seamless procedure that maintains the user data and RAID configuration.

The Smart Array B110i also includes the following-

- Support for solid state disks
- LED support
- SATA drive firmware flashing (offline)

Hot Plug technology

The HP Smart Array B110i SATA Raid Hot Plug Advance Pack provides hot-plug RAID support for the embedded SATA controller. The Hot Plug Advance Pack is available on the DL165 G7 server as a license key on cold plug and hot plug models. The existing B110i Raid 0, 1, and 1+0 will continue to support selected G6 and G7 non-hot plug and cold plug models with no fee.⁸

For more information about the B110i SATA Software RAID, download the B110i user guide at <http://h20000.www2.hp.com/bc/docs/support/SupportManual/c01706551/c01706551.pdf>

Smart Array Advanced Pack

HP Smart Array Advanced Pack (SAAP) firmware provides advanced functionality within Smart Array controllers. This firmware further enhances performance, reliability, and data availability. The Smart Array controller firmware stack supports SAAP. It can be enabled on the P212, P410, P410i, and P411 controllers.

SAAP requires a license key for activation. After activation, administrators can employ several capabilities:

- RAID 6 with Advanced Data Guarding (ADG) protects against failure of any two drives. It requires a minimum of four drives, but only two will be available for data. ADG can tolerate multiple simultaneous drive failures without downtime or data loss and is ideal for applications requiring large logical volumes because it can safely protect a single volume of up to 56 disk drives. RAID ADG also offers lower implementation costs and greater usable capacity per U than RAID 1.
- RAID 60 allows administrators to split the RAID storage across multiple external boxes. It requires a minimum of eight drives, but only four will be available for data.

⁸ Non-hot plug models cable hard drives directly from the motherboard. Cold plug models utilize a hard drive backplane but do not contain a hot plug HDD controller. These models can be upgraded to hot plug with the B110i software RAID advanced pack license, or with the introduction of a Smart Array hardware-based controller.

- Advanced Capacity Expansion (ACE) automates higher capacity migration using capacity transformation to remove logical drives by shrinking and then expanding them online. Standard drive migration and expansion remain unchanged.
- Mirror Splitting and Recombining in Offline Mode breaks a RAID 1 configuration into two RAID 0 configurations. This is similar to a scaled down rollback functionality that requires two disk drives.
- Drive Erase completely erases physical disks or logical volumes. It is useful when decommissioning, redeploying, or returning hard drives.
- Video On Demand Performance Optimization decreases latency and improves video streaming.

More information about SAAP is available at www.hp.com/go/SAAP.

NOTE:

At a minimum, a 256 MB cache and battery kit is required to enable the SAAP license key. SAAP is not available on Zero Memory Configurations.

Solid state drives

HP has introduced the second generation of solid state drives (SSD) for ProLiant servers. These solid state drives are 3 Gb/s SATA interface in both 60GB and 120GB capacities. These drives, based on NAND Single Level Cell flash technology, are implemented as SFF and LFF hot plug devices on the HP universal drive carrier for general use across the ProLiant portfolio. They deliver higher performance, lower latency, and low power solutions when compared with traditional rotating media.

You can use the HP second generation SSDs with the present generation Smart Array controllers based on the PM8011 SRC MIPS processor on select ProLiant G6 and G7 servers. See the server QuickSpecs to confirm that SSDs are supported.

For more information on HP second generation SSDs, download the “Drive technology overview” technology brief at www2.hp.com/bc/docs/support/SupportManual/c01071496/c01071496.pdf

Power and thermal technologies

HP engineers have developed a robust set of power and thermal technologies and components to manage power within ProLiant 100-series G7 servers. The following technologies improve power efficiency throughout the power delivery chain:

- Efficient power delivery
- Thermal sensors and fan control
- Processor management technologies

Administrators can disable certain components and capabilities within the ProLiant 100-series G7 servers or reduce capabilities to bring the components to a lower power state.

Efficient power delivery

With the exception of performance models, the ProLiant AMD DL165 G7 server is equipped with the high-efficiency 500W multi-output power supply that improves performance-watt ratio. This power supply is neither hot pluggable nor auto-switching. Common slot power supplies are an option in those G7 models supporting the common slot architecture, including the DL165 G7 server. To see all the power supply options offered with any HP server, consult the server QuickSpecs or go to the HP ProLiant Server Compatibility Guide at <http://www.hpproliantoptions.com/intro/>

Common Slot power supplies

The HP Common Slot power strategy provides power supply commonality across supported ProLiant G7 servers. Three different sized common slot power supplies are available so customers can choose the most effective power supply to match their needs. "Right sizing" power supplies lets you more closely match the power supply to the server power requirements in specific environments, significantly reducing wasted power. The HP Common Slot power strategy has also reduced the number of power supply designs, which in turn reduces the number of spares you need to keep in the data center.

Power supply efficiency relates to the level of effective transfer and delivery of power through the power chain. Table 2 shows that HP power supplies have achieved efficiency ratings of up to 94%, meeting the Climate Savers Platinum requirements.

Table 2. HP power supply efficiency and Climate Savers rating

Power supply	Efficiency	Rating
460W AC	up to 92% efficiency	Climate Savers Gold
750W AC	up to 92% efficiency	Climate Savers Gold
1200W AC	up to 94% efficiency	Climate Savers Platinum
1200W 48VDC	up to 90% efficiency	Climate Savers Silver

Use the HP Power Advisor to help determine which power supplies will best meet your needs: www.hp.com/go/hppoweradvisor.

Thermal sensors and fan control

Thermal sensors and fan control in the ProLiant AMD based 100-series G7 servers remain unchanged from the previous generation G6 servers. Sensor data from the processor, hard drive enclosure, and mother board provide thermal information to firmware that controls fan speed. Fan speed is altered to increase or decrease cooling based on the sensor data and location. If one fan fails, all the other fans default to high speed so the server remains within thermal specifications.

Optional redundant fans

The ProLiant DL165 G7 default configuration consists of four fans in a one-processor platform, and seven fans in a two-processor platform. The seven fan configuration provides N+1 redundancy. N+1 redundant means that if one fan fails the system can continue running until a planned downtime when the failed fan can be replaced. Customers can purchase the N+1 fan redundancy option (an additional three fans) for the one-processor platform.

Managing processor technologies

The AMD-P suite of power management technologies is built into AMD Opteron 6100 series processors. The following processor management technologies are supported in the ProLiant AMD-based 100-series G7 servers.

Cool Speed™

Cool Speed technology protects processor integrity by reducing power-states when an established temperature limit is reached. This technology allows a server to continue operating if the processor thermal environment exceeds safe operational limits.

It is important for managers to understand that Cool Speed is an automated processor feature. Within a warm room rack environment, some server platforms may initiate this process while other servers do

not. Given the same workload, Cool Speed activation will depend on the installed processor model and local environmental conditions. AMD Cool Speed is enabled by default and is not a user option.

C1E™

In G34 based-systems, the Northbridge chipset detects when all of the processor cores are idle and communicates that command to the Southbridge chipset. After the command is sent, the Northbridge and HT links are powered down and the cores go into a deeper sleep state. Depending on system configuration, this feature can equate to a significant datacenter power savings when the Northbridge and HT links are powered down and cores are at idle. The OS manages C1E and it is enabled or disabled through the BSU.

Active Processor Cores

This technology allows users to select the number of active cores per processor (minimum one, up to full number supported). This technology can improve performance of applications not designed to employ the full number of processor cores. Reducing the active core count can increase memory bandwidth per thread. Selecting the number of active cores is a BSU option.

Server management

ProLiant ML and DL 100-series G7 server users have different computing requirements. Consequently, the way in which customers manage and control servers can differ. With these requirements in mind, this section examines the following management topics:

- Systems management and monitoring
- 100i remote management and control (LO100i)
- Intelligent Platform Management Interface (IPMI) 2.0 and Data Center Management Interface (DCMI) 1.0 Standards

Some of these technologies are new tools for the ProLiant 100-series G7 servers, while others have been available with previous generations of ProLiant 100-series servers. Users may already be familiar with Agents, SmartStart Scripting Toolkit (SSSTK), and software Smart Components. These tools are now available for ProLiant 100-series G7 servers. The tools let users deploy many servers at once and manage them with HP SIM and Insight Management Agents.

Systems management and monitoring

Unplanned downtime can be significantly reduced through alerting provided by Insight Management Agents, which are based on SNMP. SNMP is the protocol developed to manage nodes (such as servers, workstations, routers, switches, and hubs) on an IP network. Network management systems learn of problems by receiving traps or change notices from network devices implementing SNMP.

Insight Management Agents

ProLiant 100-series G7 servers can use the same SNMP-based Insight Management Agents that are supported by other ProLiant servers. This means that administrators can use HP SIM 5.3 and greater to manage ProLiant 100-series G7 servers. Administrators can also use any other SNMP-based management tool. Support Automation Services are provided on 100-series G7 servers through these SNMP agents. The agents are included on the ProLiant 100-series model-specific “Easy Set-up” CDs. They are also available at www.hp.com/servers/easyssetup.

The SNMP agents enable the following capabilities on 100-series G7 servers:

- Health monitoring capabilities, including monitoring for drives, fans, network, power supplies, and temperature

- Alerting capabilities, including basic alert notification for Smart Array drive pre-failure only
- Performance monitoring capabilities providing information on processor, memory, disk free space, network utilization, and I/O

Remote management

LO100i is included in all ProLiant 100-series G7 servers. LO100i represents the core embedded management functions in ProLiant ML, DL, and SL100-series servers. It works in concert with HP Systems Insight Manager (SIM), BSU, and Option ROM Configuration for Arrays (ORCA) to provide remote management, deployment, and control functions without additional software. You can access LO100i locally through the BSU, or remotely with a web browser through HP SIM. You can add additional software functionality with the HP Lights-Out 100i Advanced Licenses which include Virtual KVM (remote graphical console) and virtual media capabilities.

HP Lights-Out 100i

HP LO100i Remote Management is hardware and firmware that provides remote server access and control capabilities through an Ethernet interface. The HP LO100i management interface is active even when the OS is not operating. The LO100i management processor is powered through the server's auxiliary power plane, so it is available as long as the server is plugged into an active power source. HP LO100i Remote Management is compatible with industry standards including IPMI 2.0 for hardware health, DCMI 1.0, and Secure Sockets Layer (SSL) and Secure Shell (SSH) technology for secure communications over public and private local area networks. HP LO100i is fully accessible using popular web browsers including Microsoft Internet Explorer 7.0 (Microsoft Windows clients), Firefox 1.5 (Windows and Linux clients), and Mozilla 1.7.12 (Windows and Linux clients). HP LO100i is also accessible using System Management Architecture for Server Hardware Command line Protocol (SMASH CLP) for Telnet and SSH sessions.

NOTE:

ProLiant 100-series G7 servers do not support LO100i Select and LO100c

LO100i Advanced Pack, available through an optional license key, contains all the features of LO100i Standard. Table 3 identifies additional functionality provided by LO100i Advanced Pack.

Table 3. LO100i functionality

Features	LO100i Standard with every ProLiant 100- series server	LO100i Advanced (license upgrade options)
Technical Support and Upgrade Licensing		Yes – G6 and later
Flatpack and electronic key delivery		Yes
Host access to IPMI environment HW status	Yes	Yes
Unencrypted browser for power, SEL, health, and key activation	Yes	Yes
Shared and dedicated network port *	Yes	Yes
DCMI 1.0	Yes – G6 and later	Yes – G6 and later
SSL & SSH security (setup in factory)	Yes – Included with G6	Yes
DNS registered names **	Yes	Yes
License manager support	Yes – G6 and later	Yes – G6 and later
IPv6 support (coexistence at launch)	Yes – G6 and later	Yes – G6 and later
Virtual KVM		License Upgrade
Virtual media (CD/DVD, floppy, ISO Image files)		License Upgrade
Power Capping		Yes – DL1000 only

* Shared and dedicated network port is currently an optional feature on some ProLiant 100-series G6 and G7 servers and may be purchased as an option.

** The DNS registered names require high speed network ports to accommodate the large packet sizes associated with this feature. High speed network ports are included on ProLiant DL series servers. The shared low speed network ports on the ProLiant ML series servers do not support DNS registered names.

LO100i includes the following feature support for the AMD 100-series G7 server platforms:

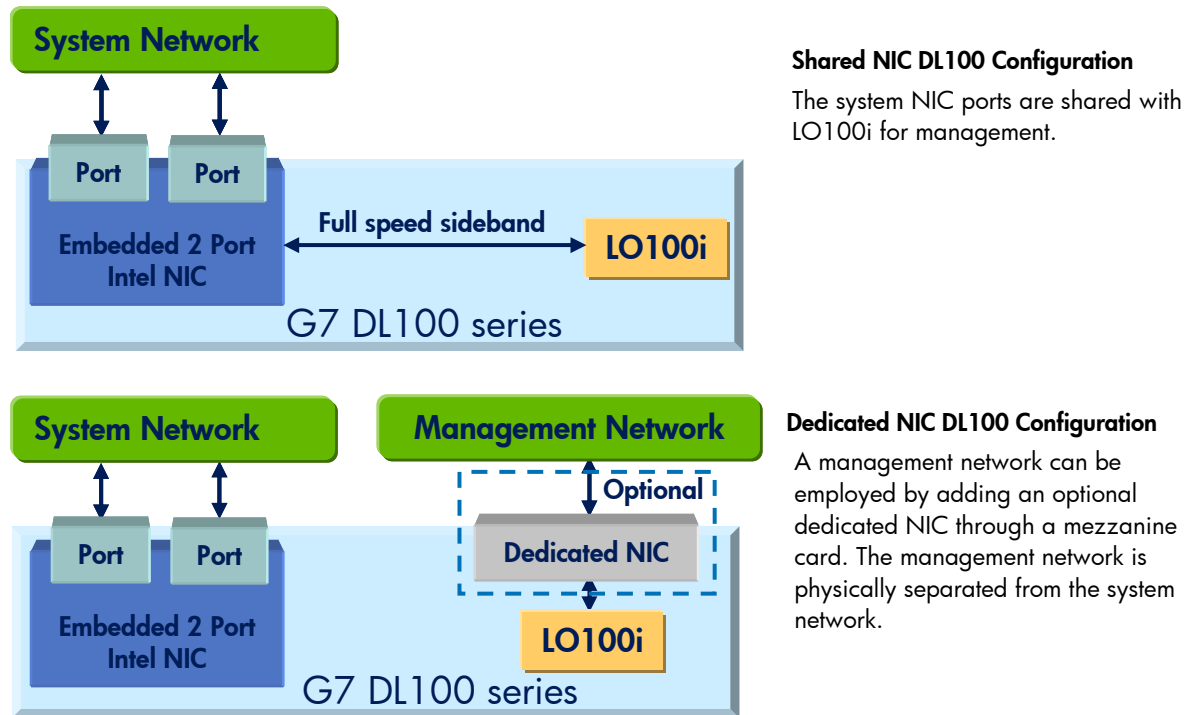
- Host access to in-band IPMI 2.0 features supported by IPMI-aware operating systems
- DNS Registration —The LO100i device comes with default host names and will automatically register with the DNS if DHCP is enabled

LO100i shared and dedicated networks

In the ProLiant DL100-series G7 servers, LO100i supports a full-speed shared Ethernet port and a dedicated Ethernet port. LO100i and the server share the full-speed Ethernet port, using the system network for both. Since the connection is full speed, it supports Graphic Remote Console and virtual media.

The dedicated Ethernet port is provided for DL 100-series G7 servers by means of an optional mezzanine card and enables a separate management network. LO100i implementation varies depending on whether the 100-series G7 server is a DL, SL, or an ML platform. Figure 3 depicts the two implementations available for the DL and SL100-series servers.

Figure 3. LO100i sideband architecture for DL and SL100-series servers



IPMI 2.0 and DCMI 1.0

ProLiant 100-series G7 servers and LO100i are architected with IPMI 2.0 and DCMI 1.0 standards so that customers in heterogeneous environments can manage these servers with either industry standard.

The following are basic compliance mandates:

- The implementation of all mandatory IPMI 2.0 and DCMI 1.0 in-band and out-of-band commands
- Reliable Local and Remote Power on/off/reset through IPMI Chassis commands
- According to IPMI 2.0, console redirection over telnet or SSH
- Server identification by device ID, globally unique identifier (GUID), asset tag, and chassis ID
- Accurate System Event Logging using IPMI
- Reliable in-band keyboard controller style (KCS) interface and out-of-band LAN interface

For more information on HP Lights-Out 100i Remote Management, go to www.hp.com/go/LO100.

Server deployment

Prior to G7, ProLiant AMD 100-series server administrators could not deploy servers using methods similar to those for ProLiant 300 series servers. The Easy Set-up CD for 100-series G7 servers now provides a user experience similar to SmartStart with deployment tools that reduce the time required for server setup and updates.

Easy Set-up CDs and their ISO images are provided with each ProLiant G7 100-series server and are used to perform assisted installations for one or two servers. Users performing multi-server installations involving large numbers of servers require the following deployment tools:

- Software Smart Components — Self-executable software consisting of driver and software
- SmartStart Scripting Toolkit (DL100s) — Automated scalable deployment utility

HP ProLiant Easy Set-up CDs

Easy Set-up CDs are available on all 100-series G7 servers and are also available as an ISO image web download. HP ProLiant Easy Set-up CDs provide easy, step-by-step, single- and multi-server utilities to streamline server setup for 100-series G7 servers. Each Easy Set-up CD includes the following:

- Boot environment and GUI
- Assisted Installation: Windows 2003 Server and Windows 2008 Server drivers
- Manual installation: Windows 2003 Server, Windows 2008 Server, and Linux drivers (specific OS support varies by server. Refer to each server's QuickSpecs for supported versions)
- HP Insight Diagnostics
- Combined Array Configuration Utility (ACU) and Array Diagnostics Utility (ADU) to offer both array controller and storage device configuration and array controller hardware testing
- SmartStart Scripting Toolkit (SSSTK) and Smart Components for software and drivers

Security

The Trusted Platform Module™ (TPM) and Microsoft® BitLocker® technology are supported in all ProLiant 100-series G6 and G7 servers by means of the available Trusted Platform Module option kit. The Trusted Platform Module v1.2 supported on ProLiant G6 and G7 servers is a microcontroller chip that can create, securely store, and manage artifacts such as passwords, certificates, and encryption keys that are used to authenticate the server platform. The TPM 1.2 chip provides a unique Endorsement Key (EK) and a unique Storage Root Key (SRK). It provides data encryption and uses RSA, SHA-1, and RNG cryptographic functions to provide access protection, OS level protection, and stolen disk protection.

The TPM 1.2 chip can also store platform measurements (hashes) to help ensure that the platform remains trustworthy. TPM enables Microsoft BitLocker, part of Windows® Server 2008. TPM is an option on all ProLiant 100-series G6 servers. For more information about TPM, go to www.hp.com/go/TPM

Microsoft BitLocker Drive Encryption (BitLocker) is a data protection feature available in Windows Server 2008. BitLocker uses the enhanced security capabilities of TPM version 1.2 to protect data and to ensure that a server running Windows Server 2008 has not been compromised while the system was offline.

OS support

HP performs extensive testing, qualification, and certification on the latest server operating systems to ensure maximum performance and reliability. HP resells and provides full service and support for Microsoft Windows operating systems, Red Hat Linux® subscriptions, Novell SUSE Linux subscriptions, Citrix XenServer, and VMware hypervisors. The latest information regarding support and deployment can be found at www.hp.com/go/ossupport.

Summary

The development and engineering innovations associated with the HP ProLiant AMD-based 100-series G7 servers help administrators decrease power costs and increase business performance. These innovations include the AMD Opteron 6100-series processor technologies, improved thermal control, common slot power supplies, DDR3 memory, and PCIe 2.0-compliant modular Smart Array controllers. The AMD 100-series G7 servers possess superior technology differentiation and management capabilities with Lights-Out 100i, SNMP management agents and the ProLiant Easy Set-up CD.

For more information

For additional information, refer to the resources listed below.

Resource description	Web address
HP Insight Management Agents for ProLiant 100 series servers	www.hp.com/go/coremanagement
HP Network Adapters for ProLiant DL and ML Servers	http://h20195.www2.hp.com/V2/GetPDF.aspx/4AA0-6064ENW.pdf
HP ProLiant DL165 G7 Server	http://h18004.www1.hp.com/products/quickspecs/13602_na/13602_na.html
HP ProLiant Onboard Administrator Powered by LO100i (remote management)	www.hp.com/go/lo100
SAS and SATA technology	www.hp.com/go/serial
HP Smart Array Advance Pack	http://h18004.www1.hp.com/products/servers/proliantstorage/arraycontrollers/smartarray-advanced/index.html
HP Smart Array controllers	www.hp.com/products/smartarray
ISS Technology Communications briefs: Memory technology evolution: an overview of system memory technologies	http://h20000.www2.hp.com/bc/docs/support/SupportManual/c00256987/c00256987.pdf
ISS Technology Communications briefs: Serial Attached SCSI storage technology brief	http://h20000.www2.hp.com/bc/docs/support/SupportManual/c01613420/c01613420.pdf

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